

There is a wide range of textbooks that cover parts of this material. Some are more advanced, and some are aimed at a lower level. Below, I try to give you a range of textbooks that you might want to refer to when you have difficulty understanding the Stochastic Differential Equations (Oksendal, B, 2003) book, the textbook that is assigned for this course.

As the course moves along, we may find other books and/or links to add to this list. So, view this as a growing document, rather than a static one.

Alternative Stochastic Calculus:

Klebaner: Introduction to Stochastic Calculus with Applications

https://www.amazon.com/Introduction-Stochastic-Calculus-Applications-3Rd/dp/1848168322/ref=sr_1_1

Financial Math Connections:

Shreve: Stochastic Calculus for Finance II: Continuous-Time Models

<https://www.amazon.com/Stochastic-Calculus-Finance-II-Continuous-Time/dp/144192311X>

Shreve: Stochastic Calculus for Finance I: The Binomial Asset Pricing Model

https://www.amazon.com/Stochastic-Calculus-Finance-Binomial-Springer/dp/0387249680/ref=pd_bxgy_14_2/133-2271060-6966351

Kuo: Introduction to Stochastic Integration:

<https://www.amazon.com/Introduction-Stochastic-Integration-Universitext-Hui-Hsiung/dp/0387287205>

Evans: An Introduction to Stochastic Differential Equations:

https://www.amazon.com/Introduction-Stochastic-Differential-Equations/dp/1470410540/ref=pd_sim_14_1/143-0362268-6915401

Karatzas and Shreve: Brownian Motion and Stochastic Calculus

https://www.amazon.com/Brownian-Stochastic-Calculus-Graduate-Mathematics/dp/0387976558/ref=sr_1_11

https://www.amazon.com/dp/144192311X/ref=sxbs_sxwds-stvpv2_3

Numerical Methods:

Kloeden and Platen: Numerical Solutions of Stochastic Differential Equations:

https://www.amazon.com/Numerical-Stochastic-Differential-Equations-Probability/dp/3540540628/ref=sr_1_5

Stochastic Processes:

Durrett: Essentials of Stochastic Processes

https://www.amazon.com/Essentials-Stochastic-Processes-Springer-Statistics/dp/331945613X/ref=sr_1_5

Collet: Discrete Stochastic Process and Applications

https://www.amazon.com/Discrete-Stochastic-Processes-Applications-Universitext/dp/3319740172/ref=sr_1_45

Probability Theory with Measures:

Kallenberg: Random Measures, Theory and Applications

https://www.amazon.com/Measures-Applications-Probability-Stochastic-Modelling/dp/3319415964/ref=sr_1_14